

DINAKARAN PRABALANATHAN

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PROFESSIONAL SUMMARY

Highly skilled Software Engineer II in Test (SE2) with over 5 years of professional experience specializing in Python-based automation frameworks, developer tools, and comprehensive test support for engineering teams at Amazon. Proven track record designing and executing functional, regression, integration, and performance test suites across complex system architectures to accelerate the software development life cycle (SDLC). Expert in collaborating with developers to establish requirements, debug intricate system failures, and build innovative engineering tools—including REST API integrations and Agentic AI-powered diagnostics—to eliminate release bottlenecks and optimize developer velocity.

CORE TECHNICAL SKILLS

- **Programming & Scripting:** Python (5+ years), Bash Scripting, JavaScript
- **Test Automation Frameworks:** PyTest (Fixtures, Parametrization, Custom Assertions), Playwright (Python & JavaScript), Selenium, tox, pexpect
- **QA Tools & Infrastructure:** Jira, TestRail, Postman, adb (Android/Linux testing)
- **Web Services & Integration:** RESTful APIs, JSON / CSV Data Pipelines
- **CI/CD & DevOps:** Git, Version Control, Dual-pipeline Deployments (Beta/Prod), AWS CDK
- **Advanced AI Tooling:** Agentic AI QA toolkits, Model Context Protocol (MCP)
- **Methodologies:** Agile/Scrum, Software Development Life Cycle (SDLC), Full-lifecycle QA, Developer Test Support

PROFESSIONAL EXPERIENCE

AMAZON | *Chennai, India*

SOFTWARE ENGINEER II (SE2) — TEST & DEVELOPER SUPPORT

July 2026 – Present

- Deliver dedicated technical test support to the core development team, collaborating with developers and project leads to establish clear testing requirements and define risk-mitigation strategies prior to production deployment.
- Review and optimize automated test infrastructure using Python, ensuring seamless integration between development branches and testing pipelines.
- Collaborated closely with firmware developers, core network groups, and project leads to establish test requirements, review peer code (85 reviews authored, 53 reviewed), and define risk mitigation strategies.

QUALITY ASSURANCE TECHNICIAN

April 2024 – July 2026

- Designed, developed, and executed comprehensive automated test scripts and frameworks using Python, pytest, and tox for continuous feature verification.
- Authored over 200+ automated test cases spanning firmware updates, device setups, sensor streaming, and media frameworks using customized fixtures and skip-lists.
- Managed the daily execution and validation of 200+ automated test suites per build cycle across nightly development and release branches to catch regressions early.
- Built a 24/7 Python automation engine that polls build systems, fetches test results via REST APIs, isolates failures, and triggers intelligent reruns without human intervention.

- Identified, documented thoroughly, and tracked over 200 critical software defects with detailed root-cause log analysis (journalctl and raw hardware scan logs), directly assisting dev teams with code fixes.
- Engineered an AI-powered QA toolkit utilizing Model Context Protocol (MCP) with 6 modular servers and 178 tools inside the IDE, reducing test analysis and regression tracking cycles from 30–50 minutes to 3–5 minutes.

DEVICE ASSOCIATE

May 2021 – April 2024

- Conducted structured **manual testing** for functional, regression, and integration phases on embedded Linux and Android hardware devices, ensuring stable product releases.
- Authored a customized Python script to automate daily hardware health checks across 2 labs and 200+ target devices, cutting evaluation hours down to minutes and auto-generating tickets in Jira.
- Developed a Branch Comparator tool to analyze test suite alignments across Mainline and Release builds, successfully reducing missing test modules during deployment readiness phases.